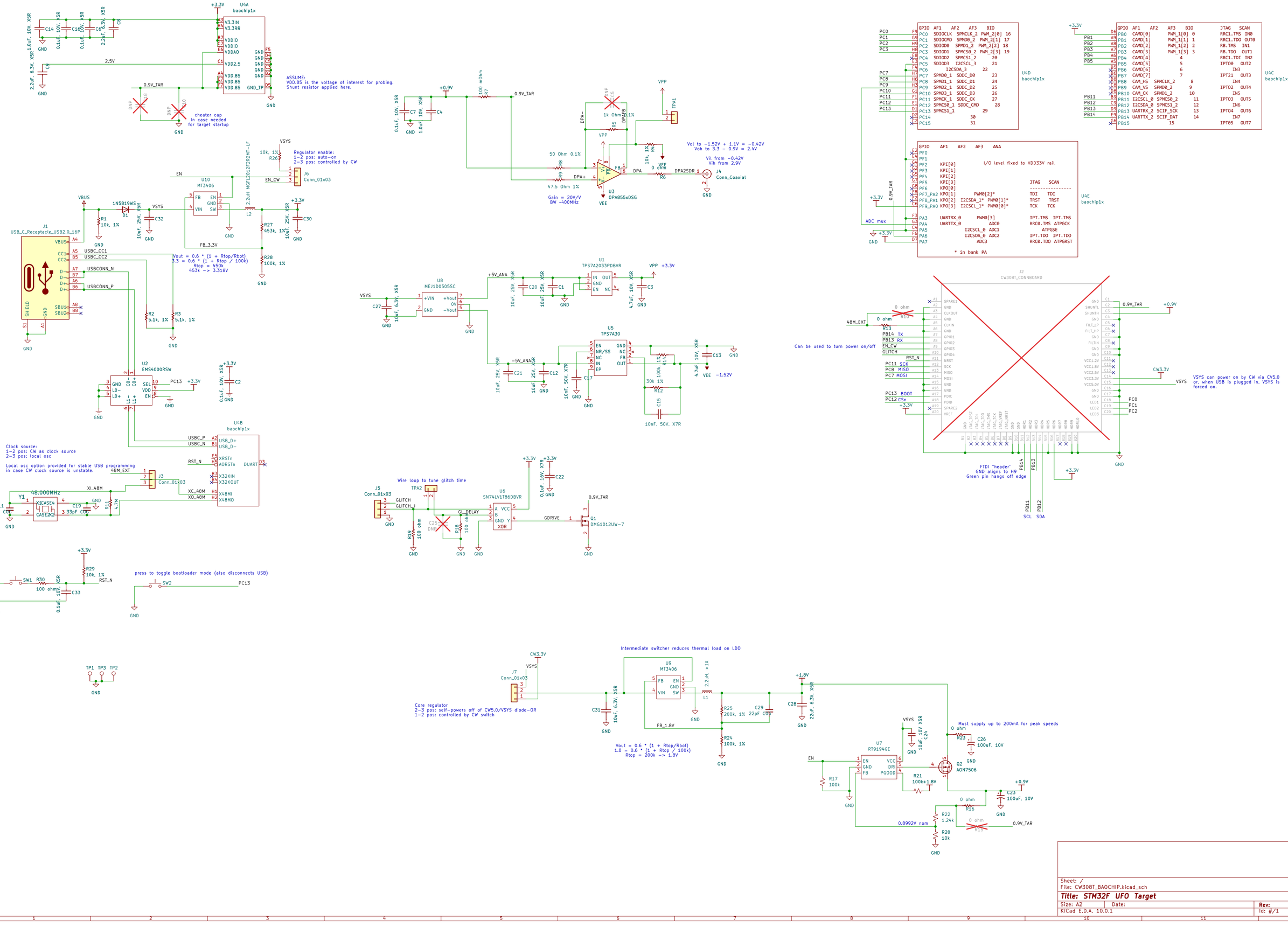

U4A
baochip1x
V3.3IN
V3.3RR
VDDIO
VDDIO
VDDAO
VDD2.5
VDD0.85
VDD0.85
VDD0.85
GND_TP
2.5V
0.9V_TAR
ASSUME:
VDD.85 is the voltage of interest for probing.
Shunt resistor applied here.
cheater cap
in case needed
for target startup
U4D
baochip1x
U4C
baochip1x
U4E
baochip1x
I/O level fixed to VDD33V rail
* in bank PA
ADC mux
U10
MT3406
1N5819WS
VBUS
VSYS
EN
EN_CW
Regulator enable:
1-2 pos: auto-on
2-3 pos: controlled by CW
J6
Conn_01x03
J1
USB_C_Receptacle_USB2.0_16P
USBC_CC1
USBC_CC2
USBCONN_N
USBCONN_P
Vout = 0.6 * (1 + Rtop/Rbot)
3.3 = 0.6 * (1 + Rtop / 100k)
Rtop = 450k
453k -> 3.318V
FB_3.3V
U2
EMS4000RSW
PC13
U8B
baochip1x
USBC_P
USBC_N
RST_N
XRSTn
AORSTn
DUART
X32KIN
X32KOUT
X48MI
X48MO
Clock source:
1-2 pos: CW as clock source
2-3 pos: local osc
Local osc option provided for stable USB programming
in case CW clock source is unstable.
48M_EXT
J3
Conn_01x03
XI_48M
XO_48M
48.000MHz
Y1
X1CASE4
CASE2X2
press to toggle bootloader mode (also disconnects USB)
SW1
SW2
TP1
TP3
TP2
+0.9V
100 mOhm
DNP
1k Ohm 1%
DPA-
DPA+
DPA_FB
50 Ohm 0.1%
47.5 Ohm 1%
Gain = 20V/V
BW ~400MHz
U3
OPA855xDSG
VPP
VEE
Vol to −1.52V + 1.1V = −0.42V
Voh to 3.3 − 0.9V = 2.4V
Vil from −0.42V
Vih from 2.9V
0 ohm
DPA2SDR
J4
Conn_Coaxial
U1
TPS7A2033PDBVR
+5V_ANA
VPP +3.3V
U8
MEJ1D0505SC
VSYS
U5
TPS7A30
−5V_ANA
−1.52V
J5
Conn_01x03
GLITCH
Wire loop to tune glitch time
U6
SN74LV1T86DBVR
GL_DELAY
GDRIVE
Q1
DMG1012UW-7
J2
CW308T_CONNBOARD
48M_EXT
Can be used to turn power on/off
PB14 TX
PB13 RX
EN_CW
GLITCH
RST_N
PC11 SCK
PC8 MISO
PC7 MOSI
PC13 BOOT
PC12 CSn
FTDI "header"
GND aligns to H9
Green pin hangs off edge
PB11 SCL
PB12 SDA
CW3.3V
VSYS can power on by CW via CV5.0
or, when USB is plugged in, VSYS is
forced on.
Core regulator
2-3 pos: self-powers off of CW5.0/VSYS diode-OR
1-2 pos: controlled by CW switch
U9
MT3406
+1.8V
Intermediate switcher reduces thermal load on LDO
Vout = 0.6 * (1 + Rtop/Rbot)
1.8 = 0.6 * (1 + Rtop / 100k)
Rtop = 200k -> 1.8V
FB_1.8V
U7
RT9194GE
Q2
AON7506
Must supply up to 200mA for peak speeds
0.8992V nom
Sheet: /
File: CW308T_BAOCHIP.kicad_sch
Title: STM32F UFO Target
Size: A2
Date:
KiCad E.D.A. 10.0.1
Rev:
Id: #/1